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Mississippi State · Ph.D. research · 2023 – present

Ultra-High-Speed PMSM Drive Development

Delay-aware multivariable control for a GaN-based PMSM drive designed for 100,000 rpm, run to 95,000 rpm on hardware.

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Problem & motivation

At 100,000 rpm the motor's electrical frequency is in the kilohertz range, and the controller gets only a few samples per electrical cycle. Computation and PWM delays then become a significant fraction of the cycle and couple the d- and q-axis currents. Conventional PI current control loses stability margin exactly where the drive needs it most.

Ultra-high-speed motors make compressors and turbomachinery smaller and lighter, but only if the drive stays stable and responsive at full speed.

Approach

- Designed FOC-based control strategies for operation up to 100,000 rpm
- Developed and analyzed MIMO control techniques that mitigate delay effects and improve high-speed stability and dynamic performance
- Validated state-of-the-art and proposed methods in real time on a PLECS RT Box FPGA model
- Implemented the algorithms as embedded code on the microcontroller and moved from simulation to physical hardware

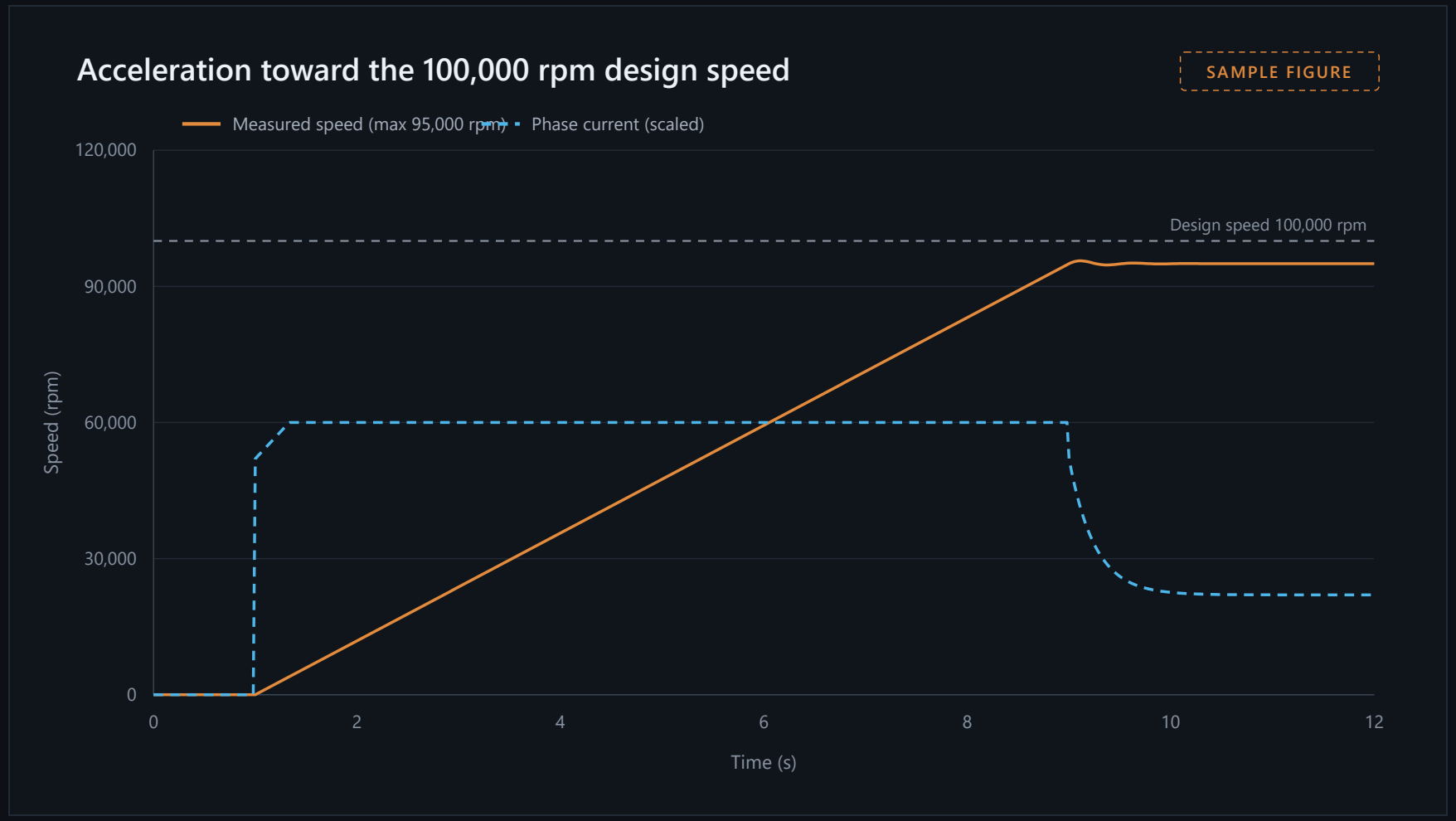
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HIL setup: PLECS RT Box, TI C2000 LaunchPad running the control firmware, and the model on screen.



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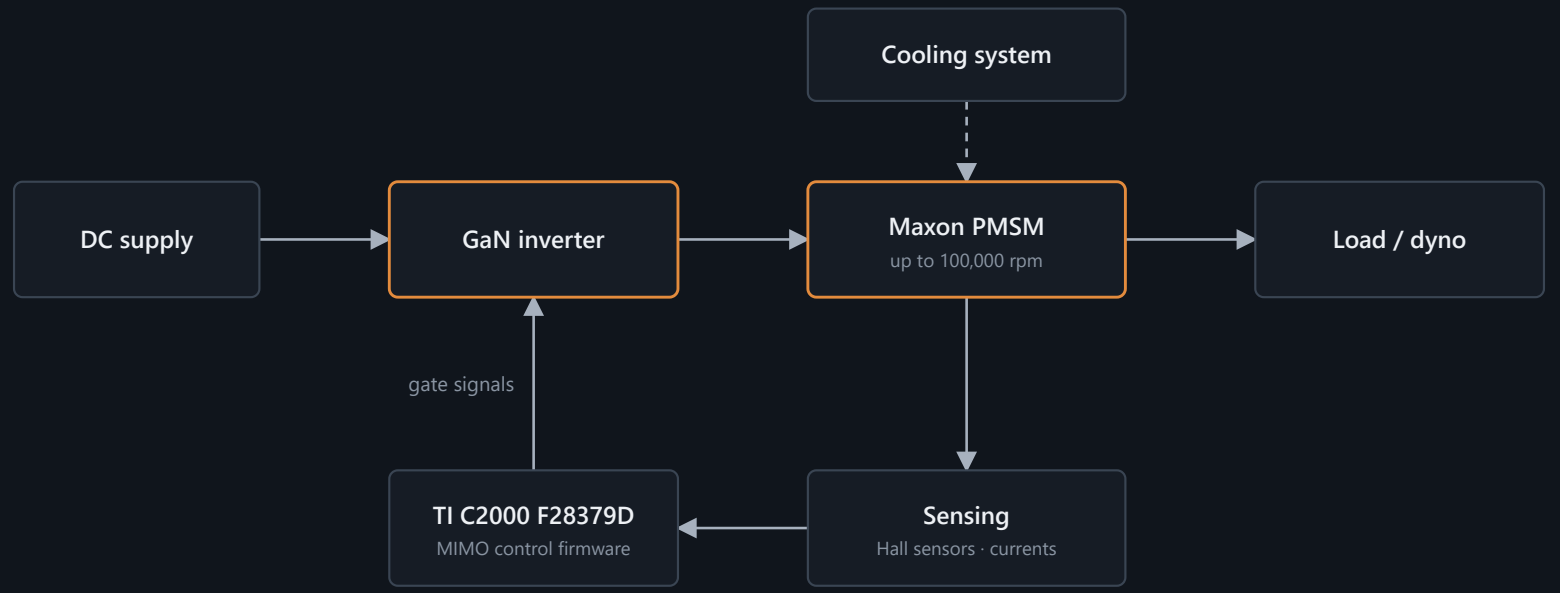
Acceleration toward the 100,000 rpm design speed; 95,000 rpm reached on hardware.



Drive system: GaN inverter, Maxon PMSM with Hall sensors, TI C2000 F28379D controller.

Drive system architecture

SAMPLE FIGURE



Results

100,000 rpm	95,000 rpm	+188%	GaN
Design speed	Reached on hardware	Speed margin (MIMO control)	Inverter technology

- Delay-aware MIMO control improved the speed margin by 188%
- Reached a maximum speed of 95,000 rpm on hardware
- Robust multivariable current control approach submitted to IEEE Transactions on Industrial Electronics (under review)

Thank you

Questions and discussion

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